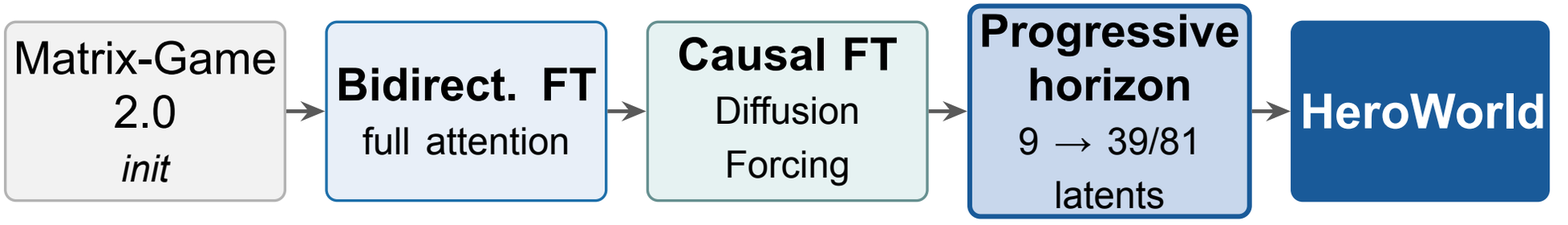
a Three-stage training recipe



b Long-horizon rollouts (64 scenes, 0–30 s)

Method	FVD-CLIP ↓ 0–10 s	FVD-CLIP ↓ 10–20 s	FVD-CLIP ↓ 20–30 s	EPE ↓	VBench ↑ Quality
SF (vanilla)	4.49	6.87	7.10	5.54	0.791
LongLive (39)	3.30	4.26	4.78	4.70	0.801
Ours (9→39)	**3.20**	**4.16**	**4.48**	**4.55**	0.794
Ours (9→81)	3.95	4.76	5.08	5.23	**0.810**

c Out-of-domain control (88 unseen scenes)

Training data	OOD action cos ↑	VBench-Quality ↑
Single domain	0.145	0.773
+ Action scaling (×3)	0.450	0.830
4 real domains	0.250	0.791
+ Simulator rollouts	**0.772** (5.3×)	**0.852**